HACK60 - AI SAMARTHYA

HCLTech × IIT Mandi

PS9: Emotion-Aware Conversational Robot System

v2.0 - Memory, Authentication, 3D Scenes & Conversational AI

Complete 60-Hour Team Workflow & Execution Plan

Start Time	End Time	Duration	Team Size
17 Apr, 1:00 AM	**19 Apr, 1:00 PM**	**60 Hours**	**4 Members**

NEW IN v2.0 - Major Enhancements

- **Multiple 3D Scenes:** Robot, environment, particles, props
- **Conversational Memory:** Remembers past interactions
- **Login System:** User profiles + history persistence
- **Robot Speaks:** Text-to-speech responses
- **Context Visualization:** 3D graph of conversation
- **Emotion-Driven Environment:** Scene changes with mood
- **Memory Graph:** Visual stored memories
- **Personalization:** Robot remembers your name

Member 1
AI + Memory Backend + LLM

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NEW FEATURES OVERVIEW (v2.0)

NEW FEATURE: Multiple 3D Models and Scenes

What: Multiple 3D models in a rich, interactive scene environment.

- **Main Robot:** Mixamo Y-Bot with 8+ animations
- **Companion Pet:** Floating drone that follows emotion
- **Environment:** Dynamic 3D room changes with emotion
- **Interactive Props:** Floating icons, particles, holographic UI
- **Memory Orbs:** Glowing 3D spheres for stored memories
- **Context Crystal:** Central 3D crystal showing context

Why it wins: Judges see fully immersive 3D world.

NEW FEATURE: Conversational Memory System

What: Robot remembers everything about user across sessions.

- **Short-term:** Last 10 interactions in session
- **Long-term:** User profile in SQLite database
- **Emotional history:** References past emotions
- **Topic tracking:** Robot knows past discussions
- **Vector embeddings:** Semantic memory retrieval
- **Personality adaptation:** Responses get personalized

Example: Session 1 user says "I love coffee". Session 2 robot says "Welcome back! Did you have your coffee today?"

NEW FEATURE: Login and User Profile System

What: Multi-user authentication with personalized interactions.

- **Sign Up:** Username, password, name, photo
- **Sign In:** JWT-based authentication
- **Profile:** Avatar, name, interests, history
- **Stats:** Total interactions, emotions, time spent
- **Robot Greeting:** Personalized welcome
- **Privacy:** Encrypted, optional anonymous mode

NEW FEATURE: Robot Speaks (Text-to-Speech)

What: Robot talks back to user.

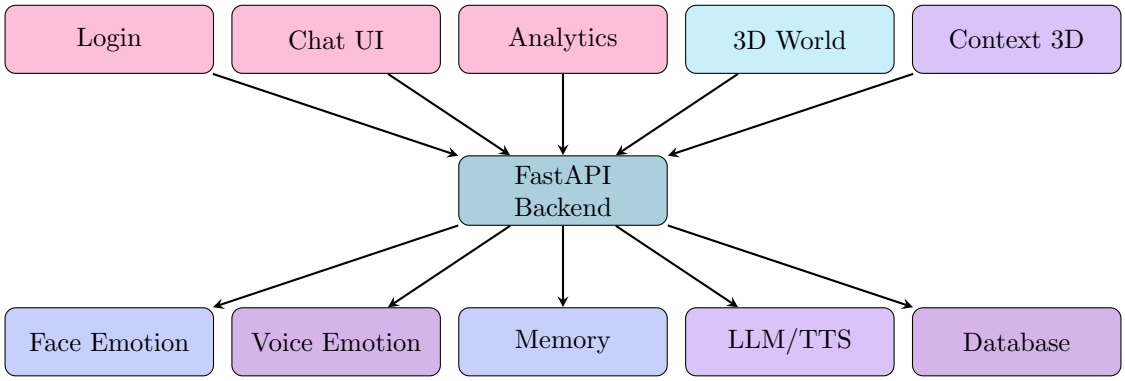
- **TTS Engine:** Web Speech API (browser-native, free)
- **Voice Selection:** Male/female options
- **Emotional Speech:** Tone changes with emotion
- **Conversational AI:** Phi-3 mini LLM
- **Natural Dialogue:** Multi-turn with context
- **Lip Sync:** Robot mouth animates while speaking

NEW FEATURE: 3D Context Exploration Visualization

What: Interactive 3D graph showing conversation context.

- **Central Node:** Current topic/emotion
- **Connected Nodes:** Related memories, topics
- **Edges:** Show relationships
- **Interactive:** Click any node to expand
- **Color Coding:** Different colors per memory type
- **Built with:** React Three Fiber + custom shaders

UPDATED SYSTEM ARCHITECTURE



Updated API Endpoints

Method	Endpoint	Purpose
POST	/api/auth/signup	Create new account
POST	/api/auth/login	Authenticate, return JWT
GET	/api/auth/profile	Get user profile
POST	/api/detect_face_emotion	Face emotion detection
POST	/api/detect_voice_emotion	Voice emotion
POST	/api/fuse_emotions	Multimodal fusion
POST	/api/chat	Get robot response
GET	/api/memory/recent	Get recent memories
GET	/api/memory/search	Semantic search
POST	/api/memory/store	Store new memory
GET	/api/context/graph	Context graph data
GET	/api/user/stats	User statistics
WS	/ws/conversation	Real-time stream

PHASE 0 - PROJECT SETUP (Hours 1-3)

DEADLINE: 17 April 2026 - 3:00 AM (All Members)

Phase 0 must complete before 3:00 AM on 17 April. Everyone confirms environment works.

Updated Dependencies

```
# Backend Setup
mkdir emotion-robot-system && cd emotion-robot-system
mkdir backend frontend models logs uploads
cd backend && python -m venv venv
source venv/bin/activate

# Core ML and AI
pip install fastapi uvicorn python-multipart websockets
pip install deepface opencv-python mediapipe
pip install torch torchvision torchaudio
pip install librosa soundfile pyaudio

# Authentication and Database
pip install sqlalchemy passlib bcrypt python-jose
pip install pydantic alembic

# Memory and Vector Store
pip install chromadb sentence-transformers
pip install langchain langchain-community

# LLM
pip install transformers accelerate

# Other
pip install python-socketio numpy pandas scikit-learn
pip install matplotlib pillow fer kaggle

# Frontend
cd ../frontend
npx create-react-app .  --template typescript
npm install @react-three/fiber @react-three/drei three
npm install @react-three/postprocessing
npm install chart.js react-chartjs-2 d3
npm install framer-motion lottie-react
npm install zustand jwt-decode axios
npm install react-router-dom
npm install tailwindcss postcss autoprefixer
npm install lucide-react react-hot-toast react-webcam
```

Folder Structure

```
emotion-robot-system/
backend/
auth/ - JWT, password, routes
memory/ - vector store, conversation, context
llm/ - chat engine, prompts
database/ - models, db connection
emotion_detection/ - face, voice
training/ - CNN training scripts
main.py - FastAPI app
frontend/
```

```
public/models/ - 3D assets
src/pages/ - Login, Signup, Dashboard, Profile
src/components/ - auth, chat, 3d, memory, context
src/hooks/ - custom hooks
src/store/ - Zustand state
```

REVISED MEMBER ROLES (v2.0)

Member 1 - AI and Memory Engineer

Primary Focus: Face emotion + Memory + LLM

- Face emotion detection (DeepFace + Custom CNN)
- **NEW:** Memory system (vector store)
- **NEW:** LLM integration for chat
- **NEW:** Context graph generation
- Grad-CAM explainability

Member 2 - Voice and Authentication Engineer

Primary Focus: Voice + Auth + Database + TTS

- Voice emotion detection (RAVDESS)
- Multimodal fusion logic
- **NEW:** JWT authentication
- **NEW:** SQLite database with profiles
- **NEW:** TTS integration
- **NEW:** Session management

Member 3 - Frontend and UX Engineer

Primary Focus: Login UI + Chat + Charts

- React app architecture
- **NEW:** Login/Signup pages
- **NEW:** Chat interface
- **NEW:** Memory timeline
- **NEW:** Profile dashboard
- Webcam component
- Analytics dashboard
- Browser TTS integration

Member 4 - 3D World Engineer

Primary Focus: ALL 3D scenes + Robot + Context

- Main robot 3D model (Mixamo)
- **NEW:** Multiple 3D models (pet, environment)
- **NEW:** Dynamic environment with emotion
- **NEW:** 3D context graph
- **NEW:** 3D memory orbs
- **NEW:** Lip sync animation
- **NEW:** Particle systems
- WebSocket integration

PHASE 1 - FOUNDATION (Hours 3-12)

DEADLINE: 17 April 2026 - 12:00 PM (All Members)

Every member must reach Phase 1 checkpoint by 12:00 PM. Integration meeting at 12:00 PM mandatory.

MEMBER 1 - Phase 1 (3:00 AM - 12:00 PM)

Task 1.1 - Face Emotion Setup (3:00 AM - 5:00 AM)

1. Build backend/emotion_detection/face_emotion.py
2. Implement FaceEmotionDetector class
3. Use DeepFace + MediaPipe
4. Test with webcam screenshot

Task 1.2 - CNN Training Background (5:00-5:30 AM)

1. Download FER-2013 from Kaggle
2. Create train_fer2013.py
3. Start training in background

Task 1.3 - Memory System Core (5:30-9:00 AM)

1. Build memory_store.py with ChromaDB
2. Use sentence-transformers (all-MiniLM-L6-v2)
3. Methods: store_memory, search_memories, get_recent
4. Build conversation.py for ConversationHistory
5. Test storing and retrieval

Task 1.4 - LLM Integration (9:00 AM - 12:00 PM)

1. Build chat_engine.py
2. Use Ollama with Phi-3 mini (or rule-based fallback)
3. Implement ChatEngine class
4. Build prompt_templates.py for each emotion
5. Add /api/chat endpoint

MEMBER 2 - Phase 1 (3:00 AM - 12:00 PM)

Task 2.1 - Voice Setup (3:00-5:00 AM)

1. Build voice_emotion.py
2. Implement VoiceEmotionDetector

3. Extract MFCC, chroma, mel features
4. Generate spectrogram as base64

Task 2.2 - RAVDESS Training (5:00-5:30 AM)

1. Download RAVDESS dataset
2. Create train_ravdess.py
3. Start training in background

Task 2.3 - Database (5:30-8:00 AM)

1. Build database/db.py with SQLAlchemy
2. Build models: User, Session, Memory, Interaction
3. Run alembic migrations
4. Test create user and query

Task 2.4 - Authentication (8:00-11:00 AM)

1. Build password.py with bcrypt
2. Build jwt_handler.py for tokens
3. Build routes.py with signup, login, profile endpoints
4. Test all endpoints with curl

Task 2.5 - TTS Endpoint (11:00 AM - 12:00 PM)

1. Add /api/tts endpoint
2. Return text and voice settings based on emotion
3. Frontend handles actual speech via Web Speech API

MEMBER 3 - Phase 1 (3:00 AM - 12:00 PM)

Task 3.1 - Routing (3:00-4:30 AM)

1. Setup React Router v6 with all routes
2. Setup Zustand store for auth state
3. Create ProtectedRoute wrapper
4. Setup Tailwind theme

Task 3.2 - Login/Signup UI (4:30-7:00 AM)

1. Build Login.tsx with gradient background
2. Glass-morphism card design
3. Show/hide password toggle
4. Loading states and error toasts

5. Build Signup.tsx with same design
6. Password strength indicator
7. Avatar upload option
8. Use Framer Motion animations

Task 3.3 - Chat Interface (7:00-10:00 AM)

1. Build ChatPanel.tsx with message list
2. Auto-scroll to bottom
3. Message bubbles user vs robot
4. Each robot message has emotion icon and speak button
5. Typing indicator animation
6. Implement Web Speech API for TTS
7. Animate message entries

Task 3.4 - Webcam (10:00 AM - 12:00 PM)

1. Build EmotionCamera.tsx
2. Capture frame every 500ms
3. Send to /api/detect_face_emotion
4. Display emotion overlay
5. Pass emotion to parent

MEMBER 4 - Phase 1 (3:00 AM - 12:00 PM)

Task 4.1 - Download All 3D Assets (3:00-5:00 AM)

Critical: Do this first!

1. **Mixamo Robot:** Y-Bot with 8 animations (Idle, Dancing, Sad, Angry, Surprised, Talking, Waving, Thinking)
2. Download as FBX, convert to GLB at gltf.report
3. Save to public/models/robot/
4. **Companion Pet:** From Sketchfab (drone or sphere)
5. **Environment Props:** Tables, holographic icons
6. **HDRi Environment:** From Polyhaven (studio.hdr)

Task 4.2 - Main 3D Scene (5:00-8:00 AM)

1. Build MainScene.tsx with React Three Fiber Canvas
2. Setup Environment with HDR
3. Add ambient, directional, spot lights

4. Add post-processing effects (Bloom)
5. Build sub-components: Robot, Pet, Floor, MemoryOrbs
6. Test all models load

Task 4.3 - Robot with Lip Sync (8:00-10:00 AM)

1. Build Robot.tsx
2. Load Mixamo Y-Bot GLB
3. Map emotions to animations
4. Smooth crossfade transitions (0.5s)
5. Implement lip sync via jaw bone oscillation
6. Add subtle floating animation
7. Add eye glow pulse

Task 4.4 - Pet and Memory Orbs (10:00 AM - 12:00 PM)

1. Build Pet.tsx - floating pet that orbits
2. Color changes with emotion
3. Trail effect using drei Trail
4. Build MemoryOrbs.tsx
5. Glowing spheres for each memory
6. Hover shows memory text as 3D billboard
7. Use instancedMesh for performance

CHECKPOINT: Phase 1 Complete - 12:00 PM on 17 April

Member 1:

- ☐ Face emotion working
- ☐ CNN training in background
- ☐ Memory store works
- ☐ LLM responds
- ☐ /api/chat working

Member 2:

- ☐ Voice extraction working
- ☐ RAVDESS training
- ☐ Database with 4 tables
- ☐ Signup/Login work
- ☐ JWT validated

Member 3:

- ☐ Routing setup
- ☐ Login UI beautiful
- ☐ Chat panel works
- ☐ TTS speaks
- ☐ Webcam captures

Member 4:

- ☐ All 3D assets ready
- ☐ Main scene renders
- ☐ Robot animations work
- ☐ Pet floats
- ☐ Memory orbs visible

INTEGRATION MEETING 1 (Hour 12)

Integration Meeting Agenda - 1 Hour

1. **(12:00 PM) M1 demos:** Face emotion + memory + LLM
2. **(12:15 PM) M2 demos:** Signup + login + database + JWT
3. **(12:30 PM) M3 demos:** Login UI + chat + TTS
4. **(12:40 PM) M4 demos:** 3D scene with robot, pet, orbs
5. **(12:50 PM) Fix bugs:** CORS, JSON, ports
6. **(1:00 PM) Merge to dev branch**

PHASE 2 - CORE BUILD (Hours 13-28)

DEADLINE: 18 April 2026 - 4:00 AM (All Members)

Phase 2 complete by 4:00 AM. Major features integrated.

MEMBER 1 - Phase 2 (1:00 PM - 4:00 AM)

Task 1.5 - Context Graph (1:00-5:00 PM)

1. Build context_graph.py
2. Generate graph data: nodes (user, memory, topic, emotion) and edges
3. Add /api/context/graph endpoint
4. Use embedding similarity for edge strength
5. Cluster related memories

Task 1.6 - Custom CNN Integration (5:00-9:00 PM)

1. Check CNN training finished
2. Integrate custom model
3. Add EmotionSmoother (5-frame average)
4. Add /api/compare_models endpoint

Task 1.7 - Memory and LLM Polish (9:00 PM - 4:00 AM)

1. Auto-extract important info from conversations
2. Importance scoring (high/medium/low)
3. Improve LLM prompts per emotion
4. Inject user name into responses
5. Reference past memories naturally
6. Memory pruning (top 100 per user)
7. Add Grad-CAM endpoint

MEMBER 2 - Phase 2 (1:00 PM - 4:00 AM)

Task 2.6 - Voice Model (1:00-5:00 PM)

1. Load trained RAVDESS model
2. Integrate into voice_emotion.py
3. Add real-time audio streaming
4. Test with various samples

Task 2.7 - User Stats (5:00-9:00 PM)

1. Build /api/user/stats endpoint
2. Return total sessions, interactions, favorite emotion
3. Build session tracking endpoints
4. Auto-end inactive sessions after 5 min
5. Log every interaction with emotion

Task 2.8 - Multimodal Fusion (9:00 PM - 1:00 AM)

1. Build fusion.py with weighted combination
2. Adaptive weights based on confidence
3. Define behavior dictionary with TTS settings per emotion

Task 2.9 - Profile Management (1:00-4:00 AM)

1. Add PUT /api/auth/profile
2. POST /api/auth/avatar for upload
3. DELETE /api/auth/account (soft delete)
4. POST /api/auth/change-password
5. User preferences (personality, notifications, privacy)

MEMBER 3 - Phase 2 (1:00 PM - 4:00 AM)

Task 3.5 - Memory Timeline (1:00-5:00 PM)

1. Build MemoryTimeline.tsx with vertical timeline
2. Cards with timestamp, emotion icon, content
3. Filter buttons (Session, Today, All Time)
4. Search bar for semantic search
5. Click memory highlights in 3D scene

Task 3.6 - Profile Dashboard (5:00-9:00 PM)

1. Build Profile.tsx

2. Large circular avatar
3. Stats cards (sessions, interactions, emotion)
4. Emotion distribution pie chart
5. Time per emotion bar chart
6. Recent sessions list
7. Edit profile modal

Task 3.7 - Voice Chat (9:00 PM - 1:00 AM)

1. Add microphone button to chat
2. Web Speech Recognition for input
3. Visual recording indicator
4. Quick reply suggestions
5. Emoji picker
6. Message reactions
7. Auto-scroll behavior

Task 3.8 - Analytics Dashboard (1:00-4:00 AM)

1. Real-time emotion radar chart
2. Emotion timeline (last 5 min)
3. Fusion panel (face + voice)
4. Session summary
5. Conversation stats
6. Export to PDF button

MEMBER 4 - Phase 2 (1:00 PM - 4:00 AM)

Task 4.5 - Dynamic Environment (1:00-6:00 PM)

1. Build EnvironmentScene.tsx
2. Different scenes per emotion
3. Smooth color transitions (lerp 1s)
4. Floor changes with emotion
5. Add 3D props (holographic icons, glowing orbs)

Task 4.6 - 3D Context Graph (6:00-11:00 PM)

This is the WOW feature!

1. Build ContextGraph3D.tsx

2. Fetch from /api/context/graph
3. Render nodes as glowing 3D spheres
4. Color by node type (user, memory, topic, emotion)
5. Animated edge lines
6. Force-directed layout (d3-force-3d)
7. Hover shows label as 3D Billboard
8. Click animates camera focus
9. Drag to reposition
10. Edge particles flowing
11. Pulse on highlighted nodes
12. Bloom for glow effect

Task 4.7 - Particle Systems (11:00 PM - 2:00 AM)

1. Build ParticleSystems.tsx
2. Use drei Sparkles or InstancedMesh
3. Different particles per emotion
4. Limit to 100 max
5. Smooth transitions on emotion change

Task 4.8 - Scene Polish (2:00-4:00 AM)

1. Combine all 3D elements
2. Post-processing (Bloom, Chromatic, Vignette, DOF)
3. Camera controls (OrbitControls, auto-rotate)
4. Camera shake on emotion change
5. Multiple preset views

PHASE 3 - INTEGRATION (Hours 28-40)

DEADLINE: 18 April 2026 - 4:00 PM

Full system integrated and tested by 4:00 PM.

PHASE 3 - ALL MEMBERS COLLABORATE

Hours 28-32 (4:00-8:00 AM) - End-to-End Integration

1. Test complete user journey:
   - Visit landing page
   - Sign up new account
   - Login with credentials
   - See main dashboard
   - Robot greets by name
   - User sends chat message
   - Robot responds with TTS
   - Robot remembers context
   - User exits, comes back
   - Robot says "Welcome back!"
2. Fix bugs in journey
3. Verify data persists

Hours 32-36 (8:00 AM - 12:00 PM) - UX Polish

1. M3+M4: Visual consistency
2. Common color palette
3. Smooth page transitions
4. Loading states everywhere
5. Error handling everywhere
6. M1+M2: Backend optimization
7. Cache LLM responses
8. Database indices
9. API response under 500ms
10. All: Cross-browser testing

Hours 36-40 (12:00-4:00 PM) - Demo Prep

1. Create 3 demo accounts:
   - Sarah - happy user, talks about coffee
   - John - mixed emotions, work topics
   - New User - empty profile
2. Pre-populate memories
3. Test demo flow with each

PHASE 4 - ENHANCEMENT (Hours 40-52)

DEADLINE: 19 April 2026 - 4:00 AM

All bonus features added. System fully polished.

MEMBER 1 - Phase 4 (4:00 PM - 4:00 AM)

1. **(4-8 PM)** Generate accuracy reports + confusion matrices
2. **(8 PM-12 AM)** Improve LLM with few-shot prompting
3. **(12-4 AM)** Documentation, README, code comments

MEMBER 2 - Phase 4 (4:00 PM - 4:00 AM)

1. **(4-8 PM)** Voice accuracy report
2. **(8 PM-12 AM)** Add features: password reset, session export
3. **(12-4 AM)** API documentation with Swagger

MEMBER 3 - Phase 4 (4:00 PM - 4:00 AM)

1. **(4-8 PM)** Final UI polish, animations, responsive
2. **(8 PM-12 AM)** Build presentation slides
3. **(12-4 AM)** Demo rehearsal coordination

MEMBER 4 - Phase 4 (4:00 PM - 4:00 AM)

1. **(4-8 PM)** Final 3D polish, performance optimization
2. **(8 PM-12 AM)** Record backup demo video
3. **(12-4 AM)** Setup demo machine

PHASE 5 - DEMO PREP (Hours 52-60)

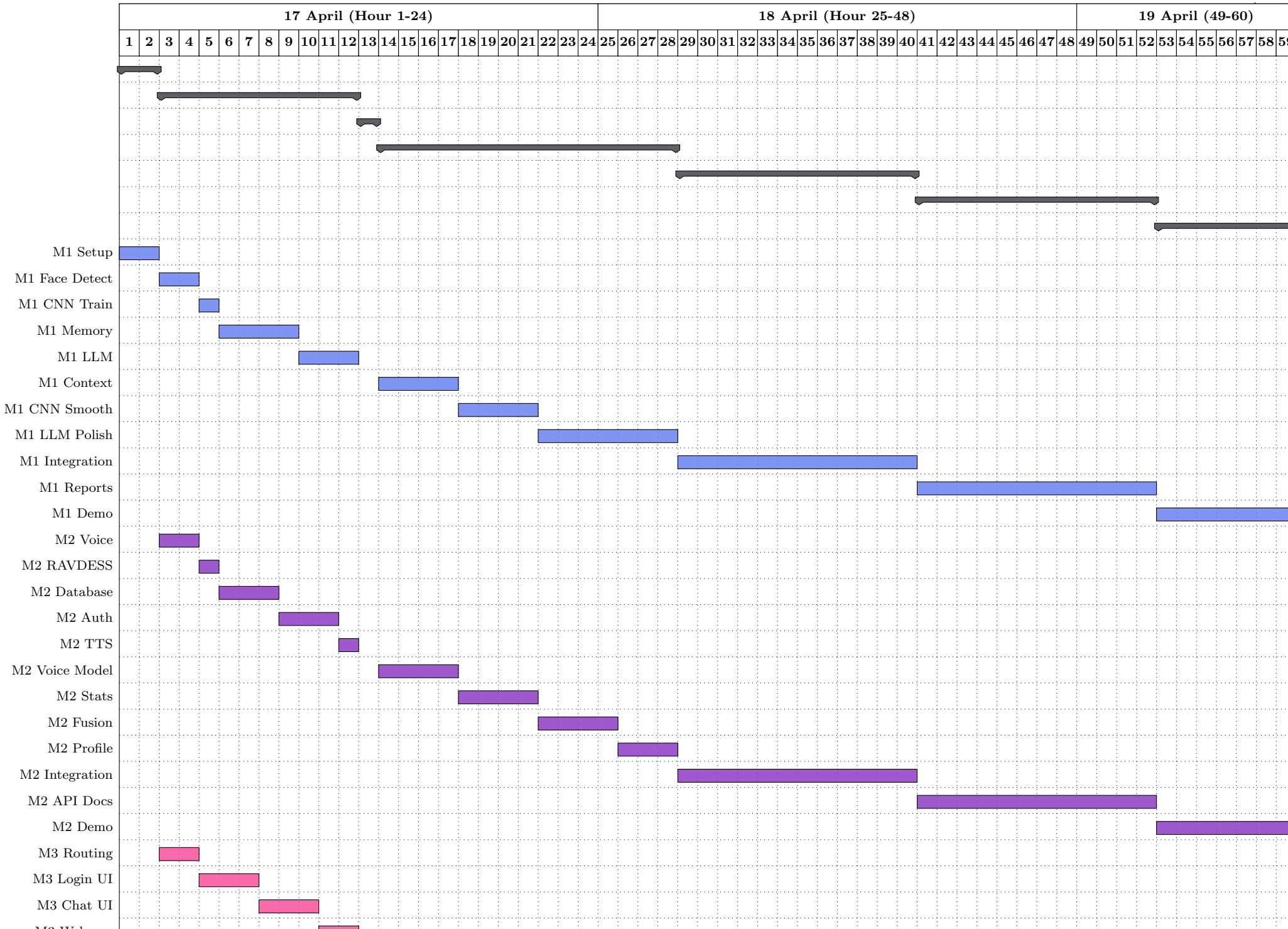
Demo Script (3 minutes)

1. **(0:00)** M1: "We built emotion-aware conversational robot with persistent memory and 3D visualization."
2. **(0:10)** M3 shows beautiful login page. M2 logs in as Sarah.
3. **(0:15)** Dashboard appears. Robot speaks: "Welcome back Sarah! I remember you mentioned coffee. How are you?"
4. **(0:25)** M3 shows full dashboard layout
5. **(0:35)** M2 smiles. Robot dances. M2 types "I'm great!" Robot speaks back.
6. **(0:55)** M1 explains 3D context graph
7. **(1:15)** M2 clicks memory orb. Camera focuses.
8. **(1:30)** M2 makes sad face. Environment dims. Robot speaks softly.
9. **(1:50)** M3 shows analytics
10. **(2:05)** M2 logs out. M1 logs in as new user.
11. **(2:25)** M4 shows technical highlights
12. **(2:45)** M1: "Future of human-robot interaction"
13. **(3:00)** INVITE JUDGE to try!

COMPLETE DEADLINE SUMMARY

Ph	Member	Task	Deadline	OK
0	All	Environment Setup	17 Apr 2:00 AM	☐
0	All	Git Repo + API Contract	17 Apr 3:00 AM	☐
1	M1	Face emotion (DeepFace)	17 Apr 5:00 AM	☐
1	M1	CNN training started	17 Apr 5:30 AM	☐
1	M1	Memory System (ChromaDB)	17 Apr 9:00 AM	☐
1	M1	LLM Integration	17 Apr 12:00 PM	☐
1	M2	Voice extraction	17 Apr 5:00 AM	☐
1	M2	RAVDESS training started	17 Apr 5:30 AM	☐
1	M2	Database & Models	17 Apr 8:00 AM	☐
1	M2	Auth System (JWT)	17 Apr 11:00 AM	☐
1	M2	TTS endpoint	17 Apr 12:00 PM	☐
1	M3	Routing + Auth state	17 Apr 4:30 AM	☐
1	M3	Login/Signup UI	17 Apr 7:00 AM	☐
1	M3	Chat interface + TTS	17 Apr 10:00 AM	☐
1	M3	Webcam + emotion display	17 Apr 12:00 PM	☐
1	M4	ALL 3D assets downloaded	17 Apr 5:00 AM	☐
1	M4	Main 3D scene setup	17 Apr 8:00 AM	☐
1	M4	Robot + lip sync	17 Apr 10:00 AM	☐
1	M4	Pet + Memory orbs	17 Apr 12:00 PM	☐
1B	All	Integration Meeting 1	17 Apr 1:00 PM	☐
2	M1	Context graph generation	17 Apr 5:00 PM	☐
2	M1	CNN integrated + smoothing	17 Apr 9:00 PM	☐
2	M1	Memory + LLM polish	18 Apr 4:00 AM	☐
2	M2	Voice model integrated	17 Apr 5:00 PM	☐
2	M2	User stats + sessions	17 Apr 9:00 PM	☐
2	M2	Multimodal fusion	18 Apr 1:00 AM	☐
2	M2	Profile management	18 Apr 4:00 AM	☐
2	M3	Memory Timeline UI	17 Apr 5:00 PM	☐
2	M3	Profile Dashboard	17 Apr 9:00 PM	☐
2	M3	Enhanced chat + voice	18 Apr 1:00 AM	☐
2	M3	Analytics Dashboard	18 Apr 4:00 AM	☐
2	M4	Dynamic environment	17 Apr 6:00 PM	☐
2	M4	3D Context Graph	17 Apr 11:00 PM	☐
2	M4	Particle systems	18 Apr 2:00 AM	☐
2	M4	Scene polish	18 Apr 4:00 AM	☐
3	All	End-to-end integration	18 Apr 8:00 AM	☐
3	All	UX Polish	18 Apr 12:00 PM	☐
3	All	Demo accounts setup	18 Apr 4:00 PM	☐
4	All	Reports + documentation	19 Apr 12:00 AM	☐
4	All	Backup video + slides	19 Apr 4:00 AM	☐
5	ALL	FINAL DEMO READY	19 Apr 1:00 PM	☐

GANTT CHART - VISUAL TIMELINE


17 April (Hour 1-24)
18 April (Hour 25-48)
19 April (49-60)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58
M1 Setup
M1 Face Detect
M1 CNN Train
M1 Memory
M1 LLM
M1 Context
M1 CNN Smooth
M1 LLM Polish
M1 Integration
M1 Reports
M1 Demo
M2 Voice
M2 RAVDESS
M2 Database
M2 Auth
M2 TTS
M2 Voice Model
M2 Stats
M2 Fusion
M2 Profile
M2 Integration
M2 API Docs
M2 Demo
M3 Routing
M3 Login UI
M3 Chat UI

Critical Path Analysis

Critical Path - Tasks That Cannot Slip

1. **M4 Asset Download (Hour 3-5):** If 3D assets not ready, M4 cannot proceed.
2. **M2 Auth System (Hour 8-11):** M3 needs auth API for login UI.
3. **M1 Memory Store (Hour 5-9):** LLM needs memory. Chat UI needs memory display.
4. **Integration Meeting 1 (Hour 13):** Anyone missing checkpoints must fix before this.
5. **End-to-End Test (Hour 28-32):** If complete journey fails, project broken.
6. **Demo Rehearsal (Hour 56-58):** Must have 3 complete run-throughs.

EMERGENCY PROTOCOLS

EMERGENCY: Backend Crashes During Demo

1. Press Ctrl+D to activate Demo Mode (pre-recorded sequence)
2. Continue narration normally
3. Meanwhile M1/M2 restart backend
4. When ready press Ctrl+D to exit demo mode
5. Backup video on USB drive if all fails

EMERGENCY: LLM Returns Bad Responses

1. Switch to rule-based templates (always ready)
2. Pre-defined responses per emotion
3. Looks just as smooth
4. Focus presentation on emotion accuracy instead

EMERGENCY: Authentication Database Fails

1. Switch to hardcoded demo accounts
2. Skip signup demo
3. Login with demo credentials directly
4. Memory still works in-memory

EMERGENCY: 3D Scene Crashes or Lags

1. Disable post-processing (Bloom, Chromatic)
2. Reduce particles to 20
3. Switch to fallback geometric robot
4. If complete failure show pre-recorded video

EMERGENCY: Webcam Not Recognized

1. Immediately play backup demo video
2. Narrate: "Let me show our recorded session"
3. Explain technical details from memory
4. Do NOT waste demo time debugging

EMERGENCY: Memory System / ChromaDB Fails

1. Switch to dict-based memory in-memory
2. Hardcode 5-10 memories for demo accounts
3. Robot still "remembers"
4. Context graph uses static data

EMERGENCY: TTS Not Working

1. Web Speech API rarely fails
2. If Chrome blocks try Firefox or Edge
3. Pre-recorded audio as final fallback
4. Robot still emotes via 3D animations

QUICK REFERENCE COMMANDS

Commands Everyone Must Know

```
 # Start Backend
cd backend && source venv/bin/activate
uvicorn main:app --reload --host 0.0.0.0 --port 8000

# Start Ollama LLM
ollama serve &
ollama run phi3:mini

# Start Frontend
cd frontend && npm start

# Start CNN Training
cd backend/training && python train_fer2013.py &

# Start Voice Training
cd backend/training && python train_ravdess.py &

# Database Migration
cd backend && alembic init alembic
alembic revision --autogenerate -m "Initial"
alembic upgrade head

# Test Authentication
curl -X POST http://localhost:8000/api/auth/signup

# Git:  Save work every 2 hours
git add -A && git commit -m "Progress"
git push origin feature/your-branch
```

Communication Protocol

- WhatsApp/Discord group - respond within 5 min during work
- Blocked? Post immediately - don't waste 20+ min alone
- Only M1 merges to main (avoid conflicts)
- Broke something? Post in group with screenshot
- Feature done? Post green checkmark with proof
- Commit format: [M1/M2/M3/M4] Phase X.Y - description
- Stuck 30+ min? Pair program - share screen
- Sleep schedule: M1+M2 sleep 11PM-7AM day 2, M3+M4 cover night

3D MODELS DOWNLOAD CHECKLIST (M4 Critical)

Complete 3D Asset Checklist - Get These by 5 AM!

1. Mixamo Character + Animations (mixamo.com)

Required animations on SAME character:

- ☐ Idle (default standing)
- ☐ Samba Dancing (for happy)
- ☐ Sad Idle (for sad)
- ☐ Angry (for angry)
- ☐ Surprised (for surprise)
- ☐ Talking 1 (for speaking)
- ☐ Waving (for greeting)
- ☐ Thinking (for processing)

2. Companion Pet (sketchfab.com - filter CC0/CC-BY)

- ☐ Robot drone OR floating sphere OR cute robot
- ☐ Lightweight (under 5MB)
- ☐ Convert to GLB format

3. Environment Models

- ☐ Floor texture (wood, concrete, futuristic)
- ☐ Background sphere or skybox
- ☐ Holographic UI panels (optional)
- ☐ Geometric shapes for memory orbs

4. HDRi Environment Maps (polyhaven.com)

- ☐ Studio (for neutral mood)
- ☐ Sunset/Warm (for happy mood)
- ☐ Night/Cool (for sad mood)
- ☐ All in HDR format

5. Sound Effects (freesound.org)

- ☐ Cheerful chime (happy detection)
- ☐ Soft tone (sad detection)
- ☐ Alert beep (angry detection)
- ☐ UI click sounds
- ☐ Notification sound

6. Conversion Tools

- ☐ gltf.report (FBX to GLB online)
- ☐ Blender (backup for complex)
- ☐ gltf-pipeline (CLI optimization)

Total Storage: 50-100MB of 3D assets

FINAL TESTING CHECKLIST

Test Case	Expected	Pass
Authentication Tests		
Sign up new user with valid data	Account created, JWT returned	☐
Login with correct credentials	JWT returned, user in	☐
Login with wrong password	Error message shown	☐
Access protected route no token	401 Unauthorized	☐
Logout clears session	Redirected to login	☐
Emotion Detection Tests		
Smile at camera	HAPPY detected over 70 percent	☐
Sad face at camera	SAD detected over 60 percent	☐
Speak in happy voice	Voice emotion HAPPY	☐
Speak in sad voice	Voice emotion SAD	☐
Fusion combines correctly	Final emotion matches	☐
Chat and Memory Tests		
Send chat message	Robot responds in 3 sec	☐
Robot speaks via TTS	Audio plays	☐
Tell robot "I love coffee"	Stored in memory	☐
Logout, login again	Robot mentions coffee	☐
Search memory "beverages"	Returns coffee memory	☐
3D Visualization Tests		
3D scene loads on dashboard	Robot, pet, env visible	☐
Robot dances when user smiles	Smooth transition	☐
Environment changes with emotion	Smooth color transitions	☐
Memory orbs visible in scene	5+ glowing spheres	☐
Context graph renders	Nodes and edges visible	☐
Click memory orb in 3D	Camera focuses on it	☐
Particle effects active	Particles match emotion	☐
Performance Tests		
Page load time	Under 3 seconds	☐
Emotion API response	Under 500ms	☐
Chat API response	Under 2 seconds	☐
3D scene FPS	Above 30 FPS	☐
No memory leaks (10 min)	Stable usage	☐
Demo Readiness Tests		
Demo accounts pre-loaded	3 users with histories	☐
Backup video recorded	MP4 on USB drive	☐
Slides finalized	PDF exported	☐
All members rehearsed	3 successful runs	☐
Demo machine ready	Power, network, browser	☐

FINAL WINNING FORMULA

WHY YOU WILL WIN

Most teams will build:
"A robot that detects emotions"

You are building:
"A robot that REMEMBERS you, GREETS you by name, TALKS back, lives in a STUNNING 3D world, and shows your CONVERSATION CONTEXT in real-time 3D"

The Difference:

- Login system - Each judge becomes a user
- Memory system - Robot remembers across sessions
- TTS - Robot SPEAKS, doesn't just emote
- Multiple 3D models - Immersive world
- Context visualization - Shows robot thinking
- Personalization - "Welcome back, Sarah!"
- Multimodal fusion - Face + Voice for accuracy

= UNFORGETTABLE DEMO

The judges will remember the team whose robot
remembered them by name and danced when they smiled.

Now go build it. You have 60 hours.

Member 1
AI & Memory
"The Brain"

Member 4
3D Worlds
"The Magician"

FINAL CHECKLIST BEFORE STARTING

- ☐ All 4 members confirmed available 60 hours
- ☐ All 4 laptops ready (16GB RAM minimum)
- ☐ Stable internet at hackathon venue
- ☐ Backup phone hotspot if WiFi fails
- ☐ WhatsApp/Discord group created
- ☐ Git repository created and shared
- ☐ Accounts ready: GitHub, Mixamo, Kaggle, Sketchfab, HuggingFace
- ☐ Power cables, mouse, headphones for everyone
- ☐ Snacks, water, energy drinks
- ☐ This document accessible to all members
- ☐ Each member has read assigned tasks
- ☐ Time synced on all machines

GOOD LUCK TEAM!

See you at the winner's podium

"Pushing limits, creating possibilities"